

● Design flow

First, I chose my $C_c = 0.2C_L = 1p$ and assumed my $GBW = 40M = \frac{g_{m1,2}}{2\pi C_c}$, so I could directly obtain $g_{m1,2} \approx 0.25m$. Next, I utilized the suggestion of referred video, setting $\frac{g_{m1,2}}{I_{d1,2}} = 16$, and then calculated $I_{d1,2} = 15.625u = I_{d3,4}$. Therefore, according to dc sweep waveform of $\frac{g_m}{I_d}$ to $\frac{I}{W}$, $\frac{I_{1,2}}{W_{1,2}} = 1.27 \rightarrow W_{1,2} = 12.3u$. Because of $I_5 = 2 \times I_{1,2} = 31.25u$, I was able to get $g_{m5} = 0.3725m$ based on $\frac{g_{m5}}{I_5} = 10$. Given that the driven current of PMOS is half as that of NMOS, PMOS's $\frac{g_m}{I_d}$ is NMOS's $\frac{g_m}{I_d}$ divided by 2. Therefore, $g_{m3,4} = 0.078m$ based on $\frac{g_{m3,4}}{I_{d3,4}} = 5$.

When it comes to P_2 , I began to give it a certain value equal to 2.5 UGB, thereby $P_2 = \frac{g_{m6}}{2\pi C_L} \times \frac{1}{1.3} = 90M \rightarrow g_{m6} \approx 3.6m$ and $I_{6,7} = 0.45mm$ by using $\frac{g_m}{I_d} = 8 \rightarrow g_{m7} = 10 \times I_6 = 4.5m$. Finally, I got the left MOS's widths by the dc sweep waveform and applied the same method to calculating the MOS of constant gm circuit.

To reduce the effect of non-dominant pole, we put a compensated resistor. What's more, it's beneficial for me to use pz analyze in ade L to get ideal R_Z ; therefore, I could effortlessly adjust the value of phase margin and unit gain bandwidth since they are inversely proportional to each other on the specific range. Moreover, I took dissipation into consideration as we will use this circuit in the future circuits, and it's obvious to see that $f_{p1} \propto g_{m1} \propto \frac{1}{C_c} \propto \frac{1}{PM} \propto UGB$, so if I changed the value of R_s to decrease my total current, the relation mentioned above is extremely pivotal to meet the specification in this homework.

Most importantly, when I commenced to achieve the lower current at first, I completely neglected the slew rate, namely $\frac{2I_{1,2}}{C_c}$. Accordingly, after realizing this issue, I changed my strategy : keep the input current unchanged and set the value of C_c as small as possible even if UGB was so close to the limitation. However, when I searched information, I found that it's inevitable to see the value of phase margin and open loop gain decline to some extent. Instead, UGB is more stable. As a result, I tried my best to maintain the stability of this circuit, and at the same time, output common voltage could highly approach 900mV, which also made it possible to obtain the small V_{os} .

● Layout consideration and Review

This time, I tended to reduce my compensate capacitance, making my layout area decrease as much as possible. However, in the process, I found that when capacitance achieved at some value, slew rate would fail to reach the specification even if I adjust the current to respond to its system. The reason behind this phenomenon might be due to the compensate capacitance is too slow so that it couldn't dominate the overall

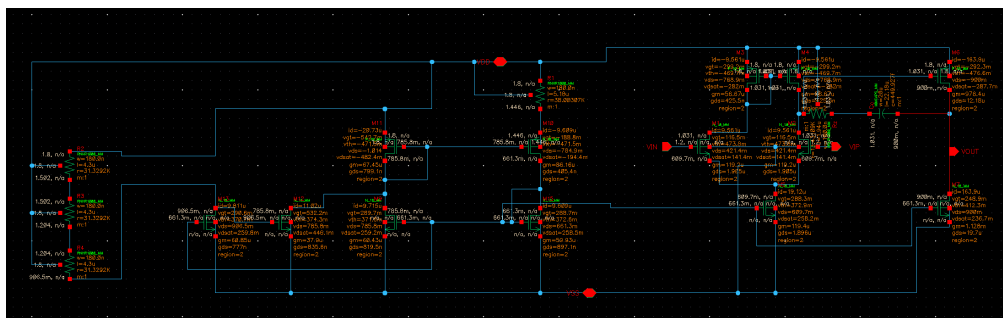
capacitors including parasite capacitors, thereby losing its partial current which should have recharge itself. Therefore, I couldn't reduce my total current excessively by dropping the value of compensate capacitance and increasing the size of Rs simultaneously.

On the other hand, my layout skills weren't mature since I didn't take resistors' size into consideration, which resulted in wasting some space. Next time, I will avoid making the same mistake and ensure that the distance between input and output can be far away to further accurate my output common voltage.

● Final Table

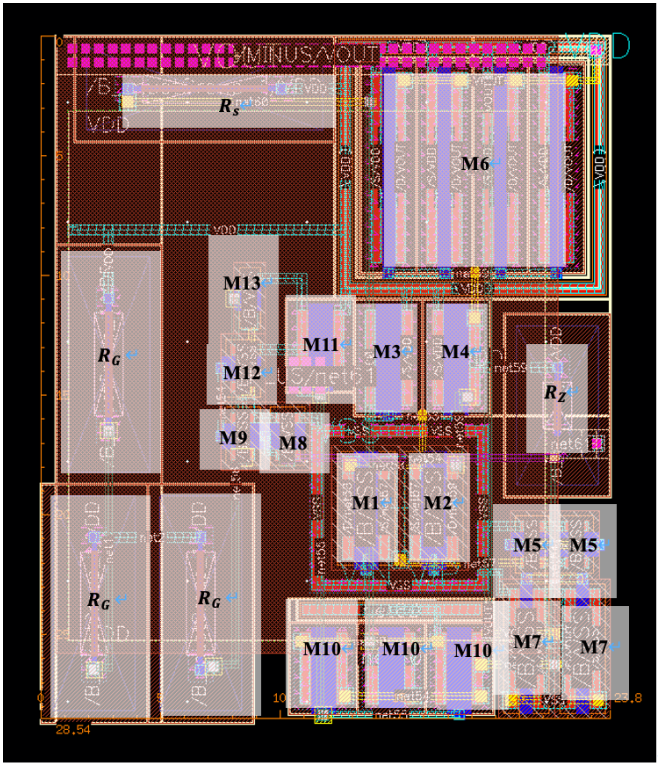
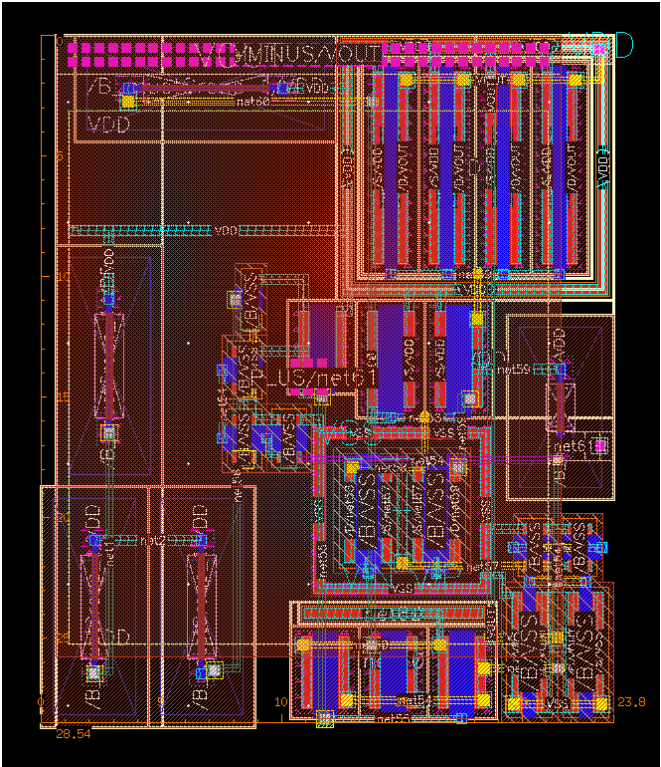
Parameters	Pre-sim	Post-sim
VDD	1.8V	1.8V
Open loop gain	63.88dB	63.91dB
CMRR	71.13dB	70.93dB
PSRR _{DD}	111.05dB	99.61dB
Input common-mode Voltage	1.2V	1.2V
Output common-mode Voltage	900mV	922.2mV
Maximum output swing(@f _{in} = 1KHz)	522.5mV~1.32V	522.6mV~1.32V
Unit-gain frequency	37.79MHz	36.86MHz
Phase Margin	66.26°	65.14°
Slew rate(rising + falling)	Rt : 43.67V/us Ft : -28.13V/us	Rt : 43.5V/us Ft : -27.45V/us
Settling time(rising + falling)	Rt : 35.99ns Ft : 63.67ns	Rt : 36.12ns Ft : 65.5ns
Input offset Voltage(MC mean abs)	12.60uV	-15.06uV
DC Power dissipation	0.40mW	0.39mW
Layout area	N/A	679.25um²

● Schematic



M_1	4.33u/1u	M_7	10u/0.5u	M_{13}	1u/1u
M_2	4.33u/1u	M_8	1u/1u	C_c	0.45p
M_3	4u/1u	M_9	1u/1u	R_z	13.4k
M_4	4u/1u	M_{10}	9u/1u	R_s	38k
M_5	2u/1u	M_{11}	3u/1u	R_G	31.2k*3=93.6k
M_6	31.92u/0.5u	M_{12}	0.37u/1u		

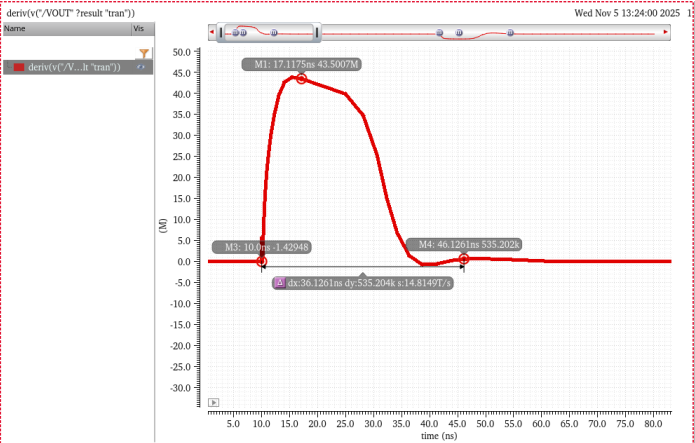
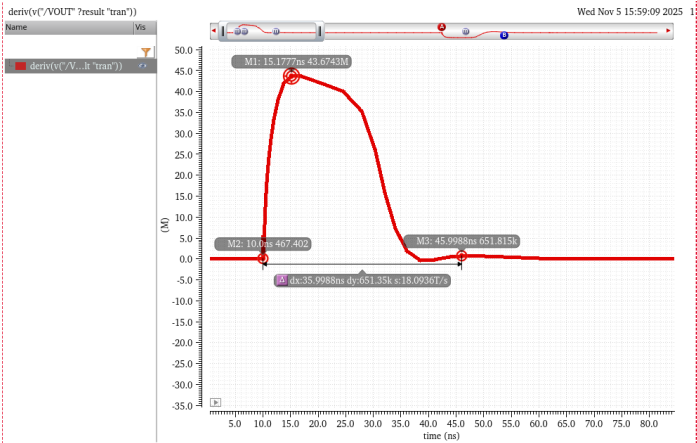
● Layout and Floorplan

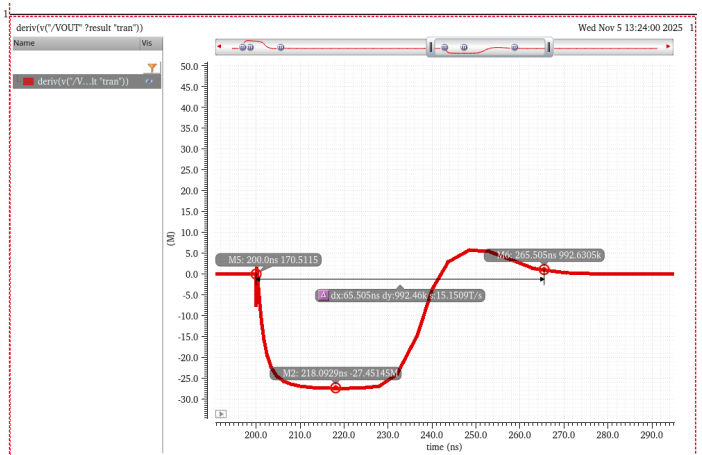
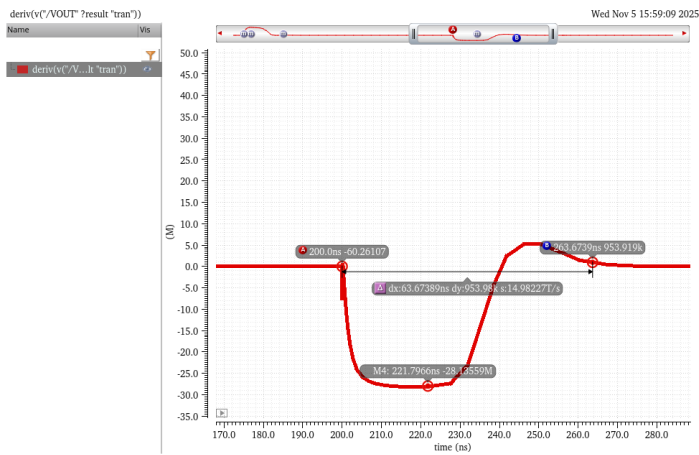


Area = $28.54 \times 23.8 \approx 679.25 \text{ (um}^2\text{)}$

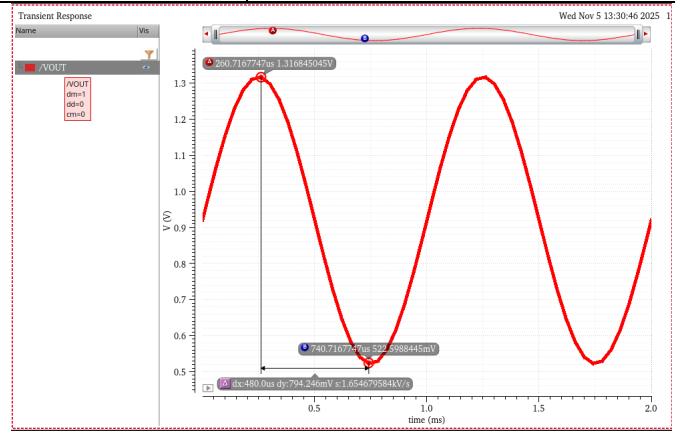
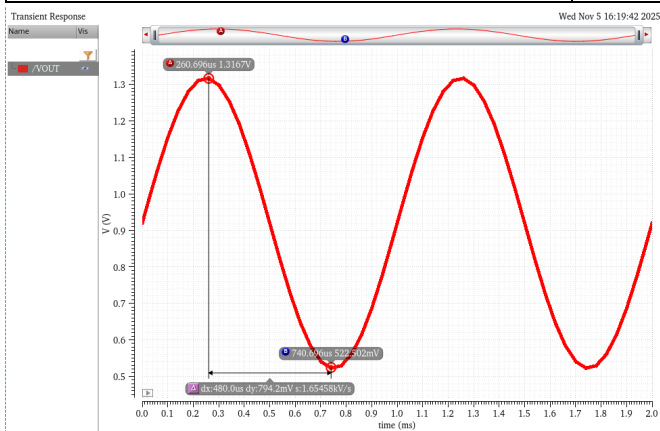
● Waveform(left: pre-sim / right: post-sim)

	Pre-sim	Post-sim
Settling time(rising/falling)	35.99ns/63.67ns	36.12ns/65.5ns
Slew rate(rising/falling)	43.67M/-28.13M	43.5M/-27.45M

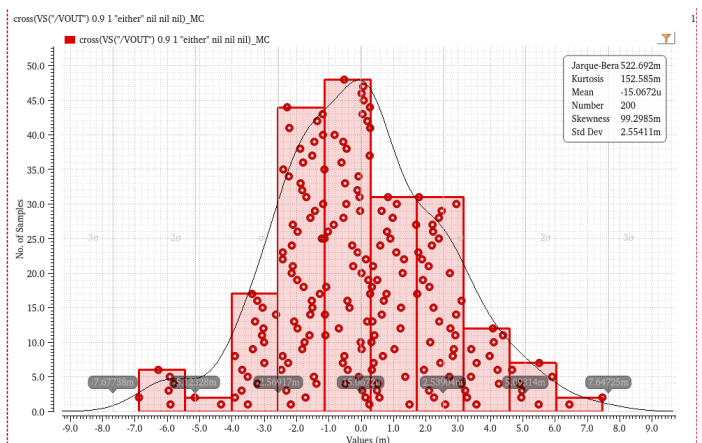
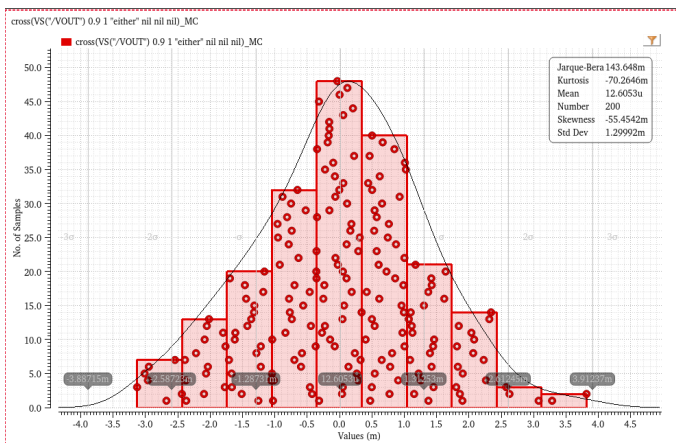




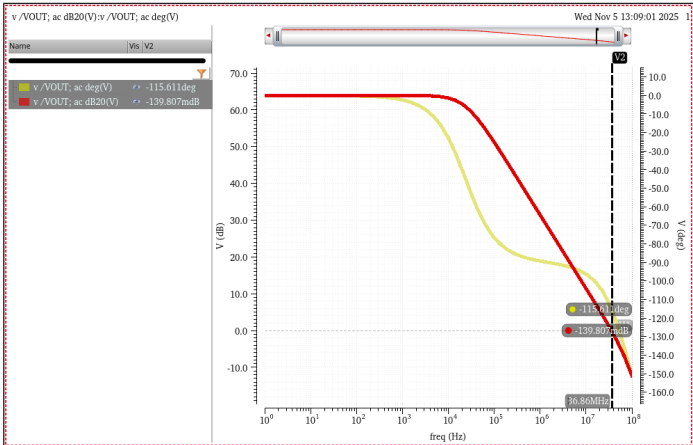
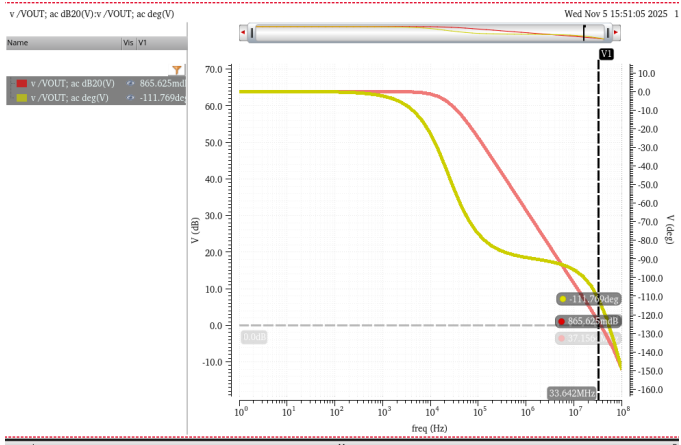
	Pre-sim	Post-sim
Maximum output swing(@ $f_{in} = 1\text{KHz}$)	522.5mV~1.32V	522.6mV~1.32V



	Pre-sim	Post-sim
Input offset Voltage(MC mean abs)	12.60uV	-15.06uV



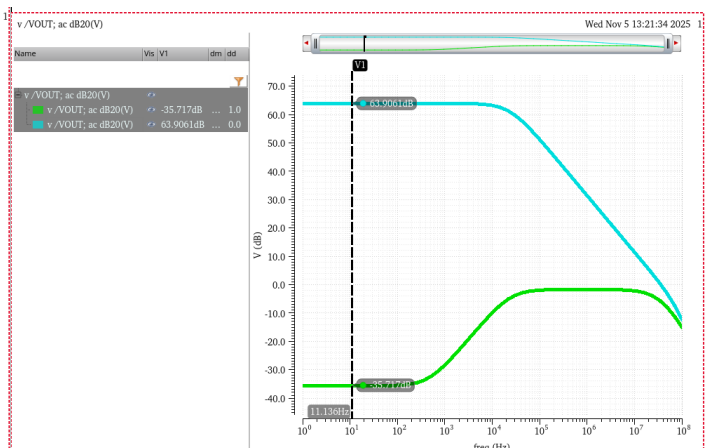
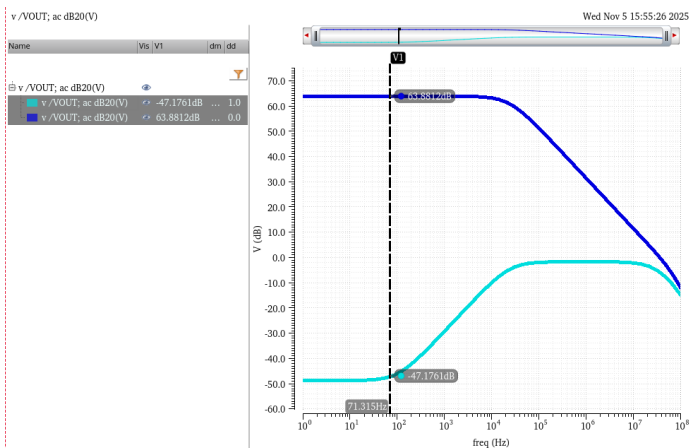
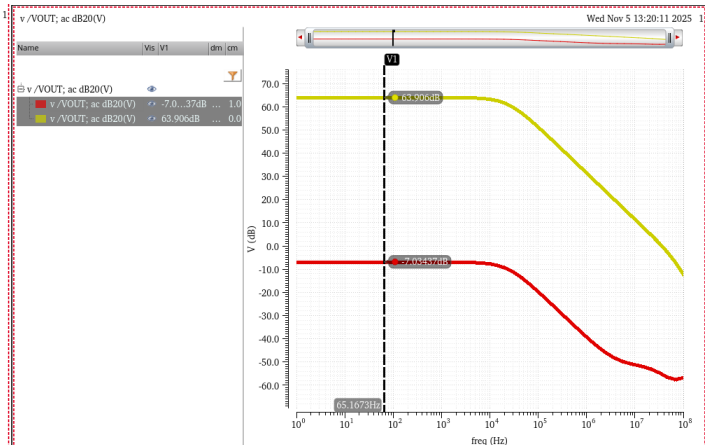
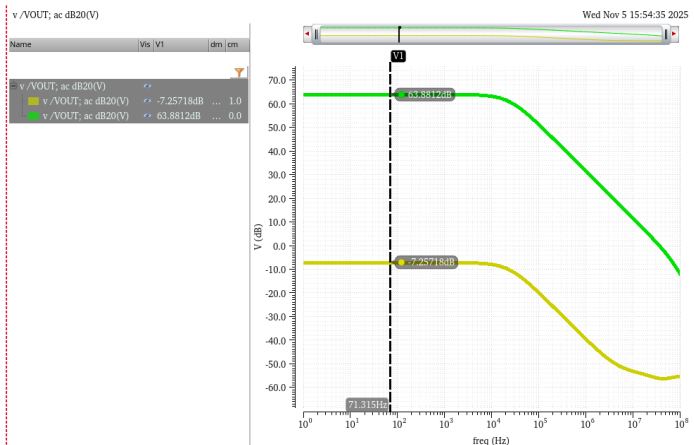
	Pre-sim	Post-sim
Open loop gain	63.88dB	63.91dB
Phase Margin	66.26°	65.14°
Unit-gain frequency	37.79MHz	36.86MHz
Output common-mode Voltage	900mV	922.2mV
I_{dc}	223.2u	221.8u



Name/Signal/Expr	Value	Plot	Save	Save Options
1 IDC("IPRB0/PLUS")	223.2u	☒	☐	
2 PM	66.26	☒	☐	
3 DC gain	63.88	☒	☐	
4 UGB	37.79M	☒	☐	
5 VDC("VOUT")	900m	☒	☐	

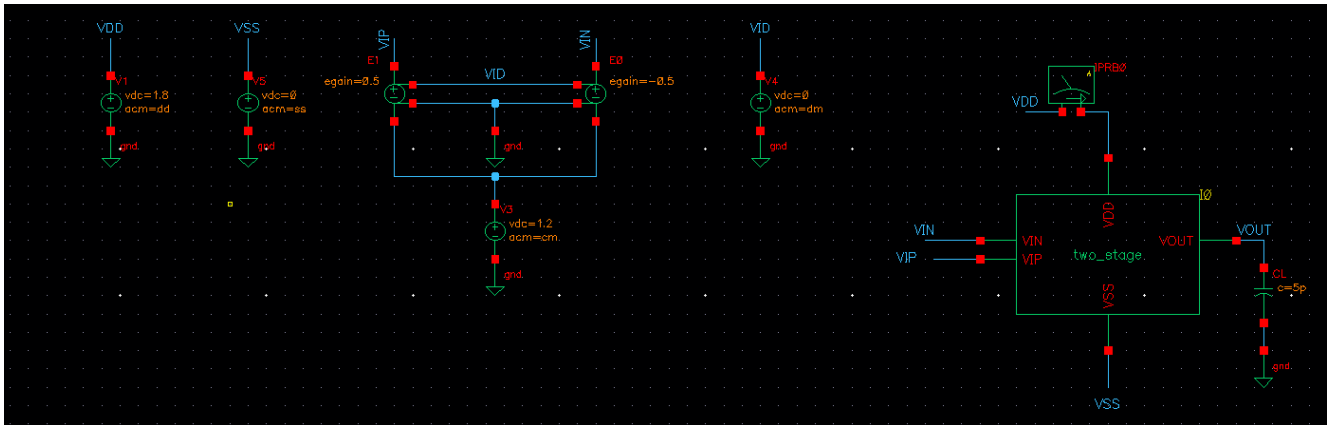
Name/Signal/Expr	Value	Plot	Save	Save Options
1 IDC("IPRB0/PLUS")	221.8u	☒	☐	
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	Pre-sim	Post-sim
CMRR	71.13dB	70.93dB
PSRR _{DD}	111.05dB	99.61dB

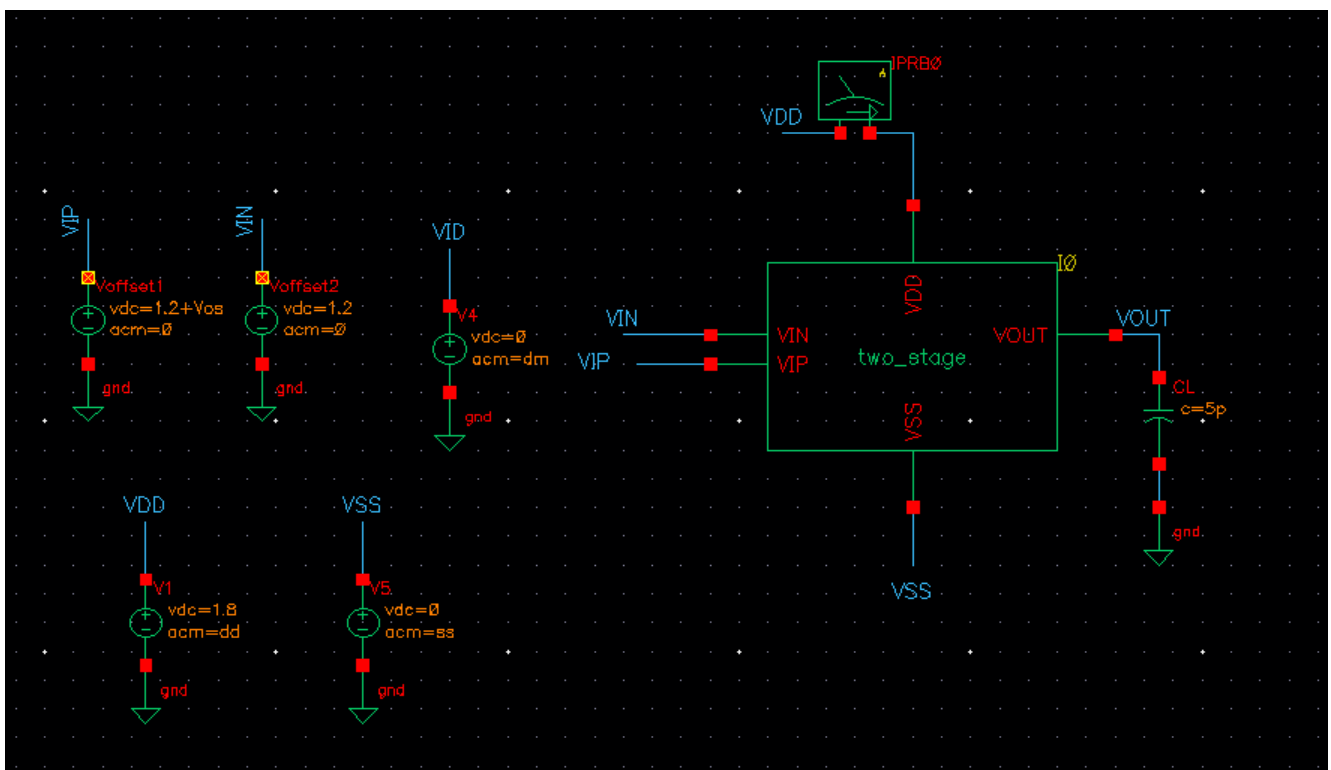


● Testbench

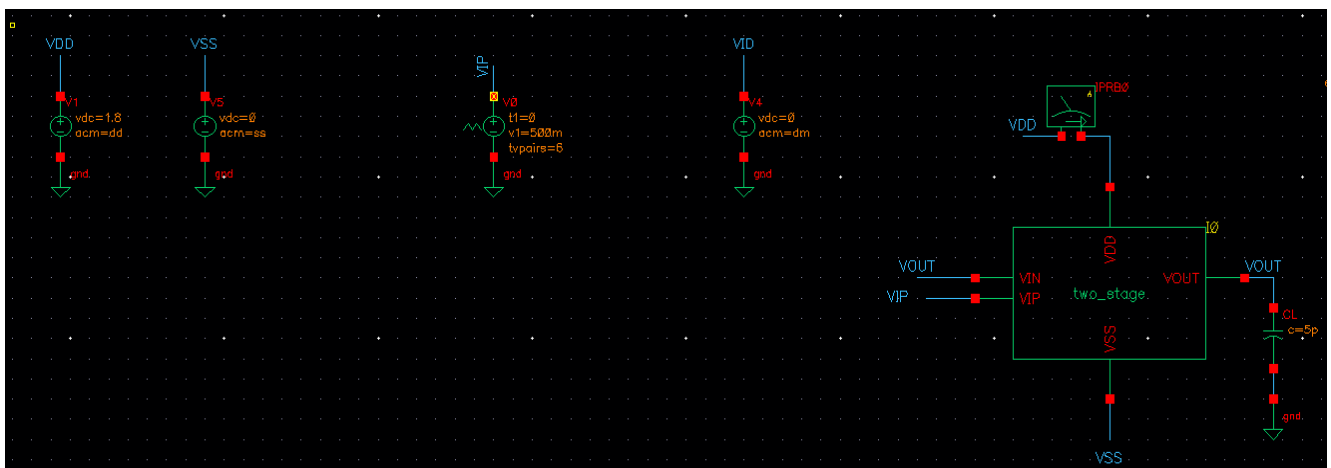
dc/ac



Input offset



Slew rate



Output swing

